

PR1 (combined) — W001-W005: Colour-line model, layer hierarchy, geometric tracing & XLSX register reconciliation

Paste this as the GitHub PR description. **Do not auto-merge** — open for engineering review only.

Branch: `pr1-w001-w005` → **Base:** `main`

Programme: Mott MacDonald / SCK CEN — MYRRHA-MINERVA Phase 1 (QCELL / RFCELL P&ID)

Toolchain: Python stdlib + `cairosvg` (PDF) + `openpyxl` (Excel) + `PyYAML`

What this PR delivers

A single consolidated foundation covering five waves:

- **W001** — source ingestion & style extraction (inline-style colour precedence).
- **W002** — colour-line decomposition into canonical process lines + validation.
- **W003** — 13 top-level layers (21 named sub-layers) hierarchy, per-element assignment.
- **W004** — geometric arrow/flow tracing, element pairing, component catalog (297), PEMO SSOT.
- **W005** — **XLSX tag & instrument register reconciliation**: cross-reference the 141 as-drawn catalog tags against the 97-tag design nomenclature register; per-TYPE coverage delta; ISA-class cross-validation; documented PPT re-allocations; merged canonical register (SSOT).

We deliberately do **not** open one PR per wave — the reviewer sees one coherent capability set.

Reproducibility (read first)

Derived outputs (`data/model/` , `data/pemo/` , `data/excel/` , `output_v6/` , `publish/` , `reports/*.xlsx`) are **git-ignored** — regenerable, not source. From a fresh clone:

```
./make.sh                                # regenerate ALL derived
outputs
PYTHONPATH=src python3 tests/test_integration_pipeline.py # source-only smoke test
(no data needed)
PYTHONPATH=src python3 tests/test_colour_model.py        # W002 assertions (after
make.sh)
PYTHONPATH=src python3 tests/test_w003_w004.py           # W003/W004 assertions
(after make.sh)
PYTHONPATH=src python3 tests/test_w005_reconciliation.py # W005 assertions (after
make.sh)
```

Verified: fresh `make.sh` exit 0 → **31/31 tests pass** → headline numbers reproduce exactly (297 components, 982→112 unmapped, pairing median 25.35 px; W005: 97 design tags vs 141 as-drawn).

Tracked source of record: `src/` , `segmentation/data/*.json` , `configs/` , `data/svg/` , `extracted/` (source reference XLSX/SVG), `tests/` , `reports/*.md` , `reports/`

W005_coverage_statistics.json ,
reports/wave_status.json , docs/ .

W005 headline finding (honest, not a defect)

The official **design nomenclature register** and the **as-drawn component catalog** use **orthogonal tag-numbering schemes**:

- **Design register** — circuit-sequential (CV001–004 , CV100/101 =40 K, TT100–111 =Pt/40 K...).
- **As-drawn catalog** — SVG-instance numbering (CV560 , TT514 , EH514 , LS-021 , HV503 ...).

→ **Exact normalized tag overlap = 0 (0.0 %)**. This is a genuine engineering finding — the two registers were authored against different conventions — **not** a pipeline defect. No matches are fabricated. Coverage is therefore measured **per instrument TYPE**, and the #1 recommended next deliverable is a **design ↔ as-drawn cross-map**.

W005 metric	Value
Design (Excel) tags / TYPES	97 / 15
As-drawn real tags / TYPES	141 / 6
As-drawn template placeholders (flagged non-reconcilable)	24 (TTxxx , EHx11 ...)
Exact tag matches	0
TYPES present in both	CV, EH, HV, PT, TT
TYPES missing from catalog	FT, FV, HX, J, LE, LI, PV, RD, SV, V (10)
As-drawn-only TYPE	LS (25 limit switches)
Documented PPT re-allocations applied	2 (TT535→PZ coldest, TT525→PZ warmest)

W005 outputs: reports/W005_XLSX_RECONCILIATION_REPORT.md , reports/W005_validation_report.md (✓ PASS), reports/W005_coverage_statistics.json , and (regenerable) data/excel/*.json , data/excel/canonical_register_v1.yaml (238-entry merged SSOT), reports/COMPONENT_CATALOG_v2.xlsx .

Capability Matrix (honest “Claim ≠ Complete” accounting)

CAN — implemented and verified

Capability	Evidence
Reduce unmapped elements via legend re-classification	982 → 112 (88.6 %)
CTM-resolved geometry extraction (bbox/centroid/shape)	geometry.py ; 7 shape classes
Pair text → components / dots / triangles / arrows → lines	315 / 205 / 100 / 132
Colour-classify text nodes	533 nodes
Assign 13 top-level layers (21 named sub-layers)	layer_assignment.json ; sum-check 2421
Render layered SVG + PDF; interactive layer-toggle atlas	QCELL 1834/14, RFCELL 591/12
Trace flow arrows + junctions	132 arrows, 36 junctions
Catalog components to Excel + HTML	297 components
Emit PEMO YAML 1.2 SSOT	122 loops, 60 heat loads
Reconcile as-drawn catalog vs design nomenclature XLSX (W005)	97 vs 141; per-TYPE delta
Detect orthogonal tag schemes (reported honestly, not forced)	0 exact overlap
Flag 10 design TYPES missing from catalog	FT/FV/HX/J/LE/LI/PV/RD/SV/V
Flag RFCELL template placeholders non-reconcilable	24 placeholders
Apply documented PPT instrument re-allocations	TT535→PZ, TT525→PZ
Emit canonical merged instrument register (SSOT)	canonical_register_v1.yaml , 238 entries

CANNOT — blocked by source data / toolchain

Capability	Reason
Map the 112 residual magenta elements to a line	No legend swatch for that colour family
Confirm true 3-D pipe routing / elevations	Source is 2-D schematic; no z-data

DID NOT / DEFERRED — possible, deferred to a later wave

Capability	Status / target wave
Design ↔ as-drawn tag cross-map (keyed on TYPE+circuit+position)	DEFERRED — W005 quantified the need
Extend as-drawn catalog to the 10 missing design TYPES	DEFERRED
Exhaustive parse of the 65 MB QSYS instrumentation PPT	DEFERRED — cost; 2 cited re-allocations encoded
Cross-drawing identity reconciliation (QCELL↔RFCELL)	DEFERRED — Wave W006
CI / GitHub Actions workflow	DEFERRED — only PLANNED

Repo hygiene note

Auto-generated office-doc preview artifacts (*_preview/ HTML renderings of .docx / .pptx) were **removed from the entire git history** and added to .gitignore . They were derived artifacts and one contained a base64 media blob that tripped GitHub secret-scanning push protection (false positive — embedded PPTX media, not a live credential). History was rewritten with git filter-repo prior to first push; commit SHAs differ from the local pre-push state.